

FedCrime: A Federated Learning Based Crime Prediction Model

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Abstract—In this thesis, we introduce FedCrime, a novel federated learning based crime prediction model that allows predicting crimes without compromising the privacy of the participants' crime incident data. A major limitation of the existing crime prediction models is the assumption of having all data at a single central server. However, in reality, people would not always feel comfortable to publicly share their data, which in turn result the loss of prediction accuracy. Traditional federated learning frameworks cannot handle the crime data sparseness problem. To address this challenge, we develop novel clustering-based data sharing and aggregation methods. Extensive experiments with real datasets show that FedCrime increases the prediction accuracy while enhancing privacy of user data.

I. INTRODUCTION

Crime prediction is crucial for crime prevention. To improve crime prediction accuracy, recent research has focused on developing deep-learning architectures that can learn from historical crime data. However, privacy concerns and delayed reporting lead to incomplete and outdated public crime data, affecting current prediction methods. In this thesis, we develop FedCrime, a federated learning-based model that protects user privacy by eliminating the need to share crime data with a centralized server.

Traditional federated learning [1] framework uses *weight sharing method*, which involves training models on clients' private data, with each client updating the model using its local data. The weights of the updated models are then aggregated on a central server to form a global model without transferring client data, thus preserving privacy. However, training crime prediction models on individual user data cannot produce a model with good prediction accuracy due to the very less crime incident data available from most users.

To address the data sparseness challenge, we propose a novel clustering based *data sharing method* within our federated learning framework, FedCrime. Users form clusters with their trusted peers and aggregate their data. Furthermore, the server's data is also divided among the clusters. This approach, combined with a novel weighted aggregation method based on user data size, enhances the accuracy of FedCrime.

II. OUR SOLUTION - FEDCRIME

Fig. 1 shows the architecture of FedCrime. The architecture consists of a central server and users. The server has the crime incident data publicly reported by the users. This dataset is incomplete. On the other hand, users store their sensitive

crime incident and harassment data on their device. Users are grouped into clusters with their trusted peers. The users of a cluster elects a cluster head and share their data with the cluster head. The central server divides the samples into equal partitions and distributes them to each cluster head. Each cluster head aggregates the received server samples with their own local data.

Cluster heads train their models using the aggregated data. The trained models from cluster heads are sent back to the server, where they are aggregated into a global model. The updated global model is then redistributed to the cluster heads for further rounds of training, continuing the iterative process.

In our existing literature, there are different aggregation methods. Two popular federated learning aggregation methods are FedAvg [1] and FedProx [2]. Both aggregate the client's model by taking a weighted sum of the models and the weights for each client are the proportion of the client sample size and total sample size. With enough data in the server, this aggregation method performs well. However, if the server data size decreases, the performance decreases. Therefore, we introduce a novel aggregation method where the server weights the client model by the proportion of the size of the client's crime data and total crime data. We are focusing on the fact that clients with more crime data will produce better models. By giving them more weight, our method will produce better-aggregated models.

III. MAIN RESULTS

A. Datasets

For experimental purposes, we generated user crime data by taking some portion of crime incidents from the Chicago Crime Dataset (2019) [6] and uniformly distributing them to users based on foursquare user check-in data used in [7].

B. Comparative Analysis

We extensively validated FedCrime (FedProx with Weighted Aggregation in Data Sharing Method) using AIST [3] as our crime prediction model. Table I compares six models on 29 region in 50% server data size and 50 cluster heads settings. In this experiment, we have shown differences between the performances of weight sharing method and data sharing method. Table I shows that model that uses data sharing method performs much better than models using weight sharing

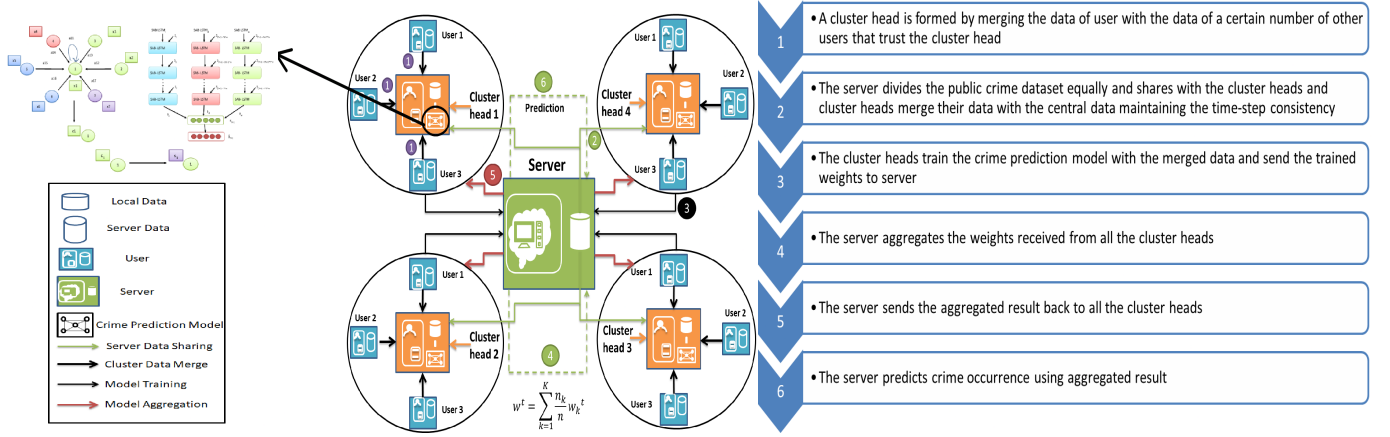


Fig. 1: Model Architecture of FedCrime

method. FedCrime outperforms the traditional Fed Avg Weight Sharing framework by 7.46% in MAE and 19.27% in MSE.

C. Effect of Varying Cluster Head

We conducted experiments across 7 regions, varying cluster head sizes (10, 25, 50, 75, 100), with 50% of crime data remaining on the server and the rest distributed among users. Fig. 2 shows that as cluster size decreases, performance improves across four model using data sharing approach, except for the traditional weight sharing approach, which remains the same for high user data sparsity. However, as cluster size decreases client privacy also decreases. We see in Fig. 2, when cluster head count ranges between 25 to 75, data sharing methods perform much better than weight sharing methods within an acceptable privacy enhancement. As a result, we can obtain both better performance and improved privacy.

D. Effect of Varying Server Data Size

We experimented across 7 regions, varying server data sizes (30%, 50%, 70%). Fig. 3 shows that our aggregation method works better than the traditional aggregation method in all server data sizes. Here, increased server data size improves model performance as expected. However, when we decrease server data, size FedCrime which uses our novel aggregating algorithm performs better than the model using traditional aggregating methods.

In summary, our model outperforms the traditional federated learning model within an acceptable range of user privacy in the clustering process. Also, with smaller server data size, our model shows better results than other models.

TABLE I: Data Sharing Method vs Weight Sharing Method

Model		Theft	
		MAE	MSE
Weight Sharing	Fed Avg	0.576	1.66
	Fed Prox	0.576	1.66
Data Sharing	Fed Avg	0.535	1.34
	Fed Prox	0.534	1.315
	Fed Avg Weighted Aggregation	0.533	1.32
	Fed Prox Weighted Aggregation	0.533	1.33

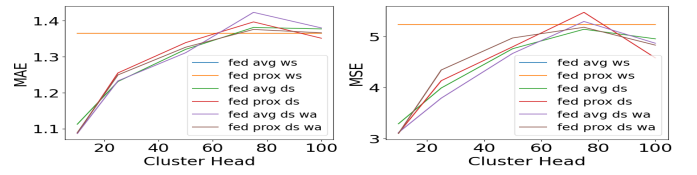


Fig. 2: Effect of varying cluster head on MAE and MSE

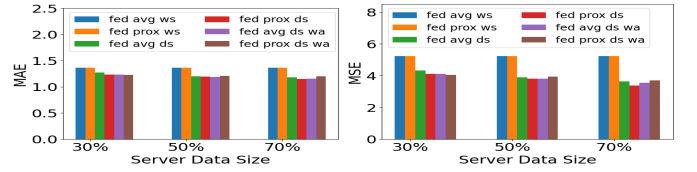


Fig. 3: Effect of varying server data size on MAE and MSE

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